| Mass | Weight |
|---|---|
| 1. It is a fundamental quantity. | It is a derived quantity. |
| 2. It has magnitude alone – scalar quantity. | It has magnitude and direction – vector quantity. |
| 3. It is the amount of matter contained in a body. | It is the normal force exerted by the surface on the object against gravitational pull. |
| 4. Remains the same everywhere. | Varies from place to place. |
| 5. It is measured using physical balance. | It is measured using spring balance. |
| 6. Its unit is kilogram. | Its unit is newton. |

## 1.9 Accuracy in Measurements

When measuring physical quantities, accuracy is important. Accuracy represents how close a measurement comes to a true value. Accuracy in measurement is center in engineering, physics and all branches of science. It is also important in our daily life. You might have seen in jewellery shops how accurately they measure gold. What will happen if a little more salt is added to food while cooking? So, it is important to be accurate when taking measurements.

Faulty instruments and human error can lead to inaccurate values. In order to get accurate values of measurement, it is always important to check the correctness of the measuring instruments. Also, repeating the measurement and getting the average value can correct the errors and give us accurate value of the measured quantity.

### Points to Remember

- Quantities which cannot be expressed in terms of any other physical quantities are called fundamental quantities. Example: Length, mass, time, temperature etc.
- Quantities which can be expressed in terms of fundamental quantities are called derived quantities. Example: Area, volume and density etc.
- A unit is the fundamental quantity with which unknown quantities are compared.
- Length, mass, time, temperature, electric current, intensity and mole are the fundamental units in SI system.
- To find the length or thickness of smaller dimensions Vernier caliper and Screw gauge are used.
- Austronomical unit is the mean distance of the Sun from the center of the Earth. $1AU=1.496 \times 10^{11}m$.
- Light year is the distance travelled by light in one year in vacuum. 1 Light year = $9.46 \times 10^{15}m$.
- Parsec is the unit of distance used to measure astronomical objects outside the solar system.
- 1 Angstrom (A°) = $10^{-10}$ m.
- SI Unit of volume is cubic metre or $m^3$. Generally volume is represented in litre (l). $1ml=1cm^3$.
- Least count of screw gauge is 0.01 mm. Lease count of Vernier caliper is 0.01 cm.
- Common balance can measure mass accurately upto 5 g.
- Accuracy of physical balance is 10 mg.



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